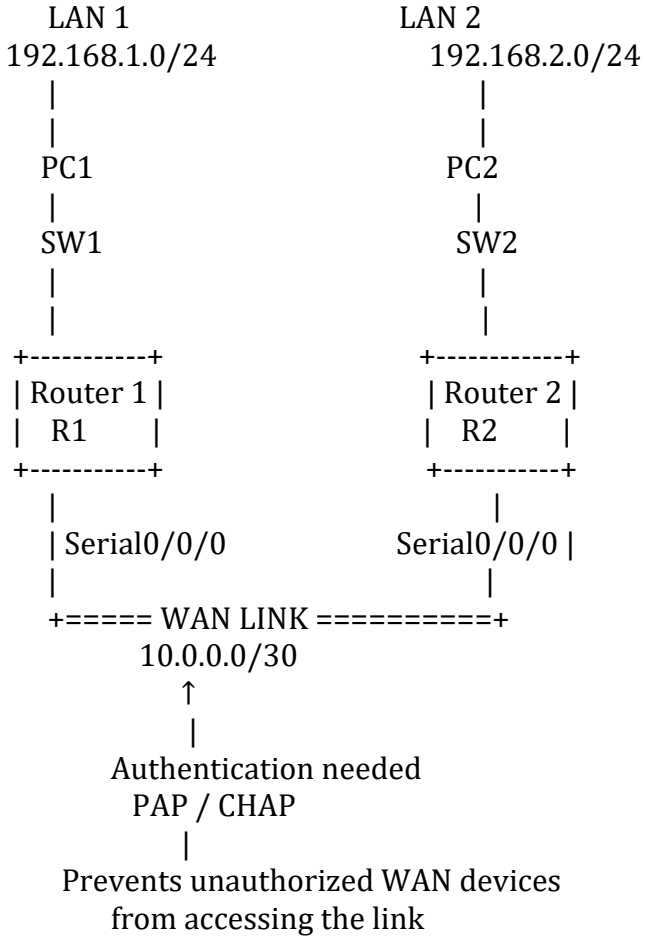
DAY - 2

WAN Authentication Protocols: Authentication is Needed in WAN, WAN Authentication Protocols: HDLC, PPP Authentication (PAP, CHAP, MS-CHAP).

1. Draw a simple network diagram showing two routers connected over a WAN link and label where authentication is needed to prevent unauthorized access.

ANS :

Simple WAN Network Diagram



Explanation

Authentication is needed on the **WAN link between Router R1 and Router R2**.

Before allowing communication, the routers can verify each other's identity using an authentication protocol such as **PAP or CHAP**.

HARSH

2. Configure a sample HDLC WAN connection between two routers using Packet Tracer or any network simulator, and explain why HDLC does not provide authentication Hint : Focus on the commands used and what is missing in terms of security.

ANS :

Network Addressing

R1 Serial0/0/0 = 10.0.0.1/30

R2 Serial0/0/0 = 10.0.0.2/30

Subnet mask:

255.255.255.252

Step 1: Configure Router R1

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# encapsulation hdlc
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# end
```

The clock rate command should normally be configured only on the **DCE side** of a back-to-back serial connection.

Step 2: Configure Router R2

```
Router# configure terminal
Router(config)# hostname R2
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# encapsulation hdlc
R2(config-if)# no shutdown
```

Step 3: Verify the Connection

On R1:

R1# show ip interface brief

Expected result:

Interface	IP-Address	Status	Protocol
Serial0/0/0	10.0.0.1	up	up

Test connectivity:

R1# ping 10.0.0.2

Successful result:

Success rate is 100 percent

Why HDLC Does Not Provide Authentication

HDLC is mainly a **Layer 2 encapsulation protocol** used to transport data across serial point-to-point links.

The important configuration commands were:

encapsulation hdlc
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown

Notice that we did **not** configure:

username
password
ppp authentication

There is no command such as:

hdlc authentication

in this basic Cisco HDLC configuration.

Therefore, the routers establish the HDLC WAN connection without checking a username or password.

Security Limitation

HDLC

|

+--- Provides framing/encapsulation

|

+--- Provides WAN communication

|

+--- Does NOT provide PAP authentication

|

+--- Does NOT provide CHAP authentication

|

+--- Does NOT verify the remote router with a username/password

For this reason, when peer authentication is required on a classic serial point-to-point link, **PPP is more suitable than HDLC** because PPP supports authentication mechanisms. Cisco documents PPP authentication support for PAP, CHAP, and MS-CHAP.

3. Set up a PPP connection between two routers in Packet Tracer and enable PAP (Password Authentication Protocol) authentication. Show the configuration commands and explain what happens if the password is incorrect.

ANS :

PAP Configuration

For this example:

R1 = 10.0.0.1/30

R2 = 10.0.0.2/30

Username R1

Username R2

Password = WANpass123

Router R1 Configuration

```
R1> enable
R1# configure terminal
R1(config)# username R2 password WANpass123
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication pap
R1(config-if)# ppp pap sent-username R1 password WANpass123
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# end
```

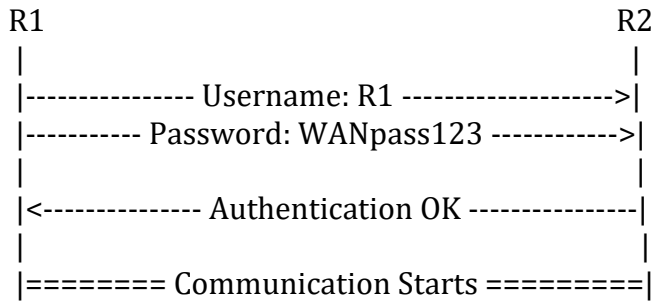
Router R2 Configuration

```
R2> enable
R2# configure terminal
R2(config)# username R1 password WANpass123
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication pap
R2(config-if)# ppp pap sent-username R2 password WANpass123
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# end
```

How PAP Authentication Works

PAP uses a simple authentication process.

PPP PAP Authentication



PAP uses a simple two-way authentication procedure and sends the username/password information without the challenge-response protection provided by CHAP. RFC 1334 describes PAP as a weak authentication method because the password is transmitted over the link in clear form.

What Happens if the PAP Password Is Incorrect?

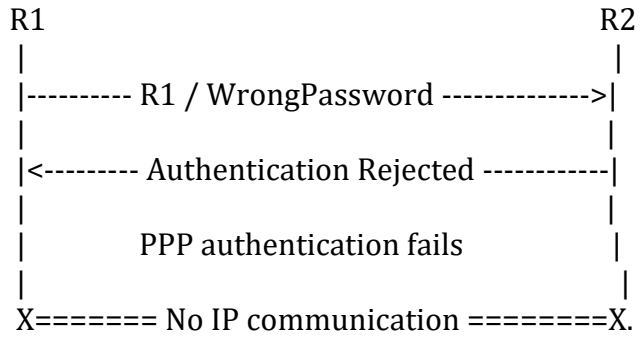
Suppose R1 sends : Username: R1

Password : WrongPassword

but R2 has : username R1 password WANpass123

The passwords do not match.

The process becomes:



The PPP authentication phase is not completed successfully, so normal network-layer communication cannot be established across the PPP connection.

4. Modify your PPP configuration to use CHAP (Challenge Handshake Authentication Protocol) instead of PAP, and describe one key difference in how CHAP improves security over PAP.

ANS :

CHAP Configuration

For CHAP, the router hostname and the username configured on its peer are important.

R1 hostname = R1

R2 must contain username R1

R2 hostname = R2

R1 must contain username R2

Both sides will use : WANchap123

as the shared CHAP secret.

Router R1 Configuration

```
R1> enable
R1# configure terminal
R1(config)# hostname R1
R1(config)# username R2 password WANchap123
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication chap
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# end
```

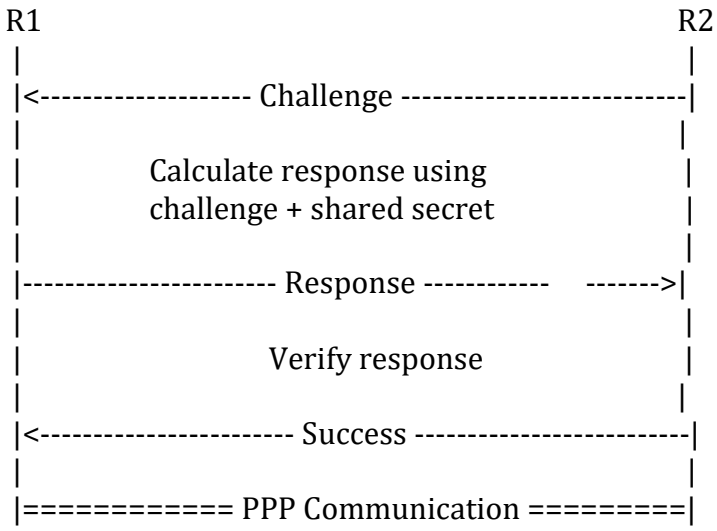
Router R2 Configuration

```
R2> enable
R2# configure terminal
R2(config)# hostname R2
R2(config)# username R1 password WANchap123
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication chap
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# end
```

Cisco documents ppp authentication chap as the command used to require CHAP authentication of a remote PPP peer.

How CHAP Works

A simplified CHAP exchange is:



The shared password itself is not transmitted directly across the link.

Key Difference Between PAP and CHAP

Feature	**PAP**	**CHAP**
Full Form	Password Authentication Protocol	Challenge Handshake Authentication Protocol
Authentication	Username + password	Challenge-response
Password sent directly	Yes	No
Handshake	Two-way	Three-way challenge-response
Security	Lower	Better than PAP
Re-authentication	Normally initial authentication	Can periodically verify peer
Cisco PPP command	ppp authentication pap	ppp authentication chap

5. Research and list two real-world scenarios (such as public WiFi login or mobile data authentication) where WAN authentication protocols like PAP, CHAP, or MS-CHAP are used, and explain why authentication is critical in each case.

ANS :

Scenario 1: ISP Broadband Using PPPoE

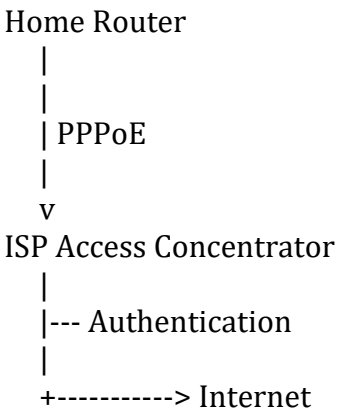
A common real-world application of PPP authentication is an Internet connection that uses **PPPoE — Point-to-Point Protocol over Ethernet**.

A customer's router may have ISP credentials similar to:

Username: customer123@isp

Password: ********

The connection can conceptually work as follows:



PPP authentication methods such as PAP can be configured in PPPoE deployments; Cisco's PPPoE documentation specifically includes PAP authentication and the ppp pap sent-username configuration.

Why Authentication Is Critical

Authentication allows the ISP to verify that the person or router requesting Internet connectivity is an authorized subscriber.

Without authentication, an unauthorized customer could attempt to use the provider's network without valid credentials.

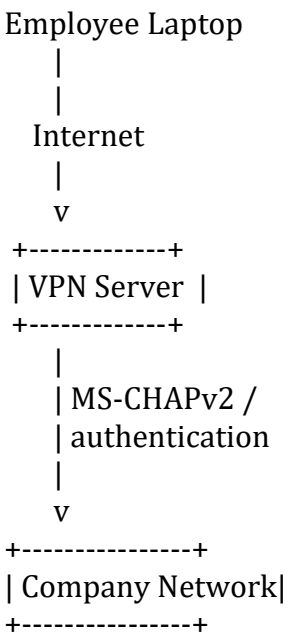
Authentication can therefore help the ISP:

- Verify subscribers.
- Control network access.
- Associate a connection with the correct customer.
- Apply the appropriate account or service policy.
- Prevent unauthorized use.

Scenario 2: Remote-Access VPN Using MS-CHAPv2

Another real-world example is a **remote employee connecting to an organization's network through a VPN**.

Example:



The employee provides authentication credentials before access to the private company network is granted.

Microsoft documents MS-CHAPv2 as an authentication option for Windows VPN and Network Policy Server configurations.

Why Authentication Is Critical

A company network may contain sensitive resources such as:

Company files

Internal applications

Servers

Databases

Authentication is therefore required to verify that the remote person is an authorized user before allowing access.

Without authentication, unauthorized Internet users could attempt to access internal company resources.

In modern deployments, stronger methods such as certificate-based EAP are often preferred when available, but MS-CHAPv2 remains relevant for understanding legacy and compatibility-oriented remote-access authentication.

Final Comparison

Protocol	Purpose	Authentication	Security
HDLC	WAN Layer 2 encapsulation	No built-in PAP/CHAP authentication	Limited
PPP	WAN point-to-point communication	Supports authentication	Depends on authentication method
PAP	PPP authentication	Username and password	Weak
CHAP	PPP authentication	Challenge-response	Better than PAP
MS-CHAP	Microsoft PPP/remote-access authentication	Microsoft challenge-response	Legacy/compatibility use
MS-CHAPv2	Improved Microsoft authentication	Challenge-response	Improved over original MS-CHAP, but stronger modern methods are preferred